

RESEARCH REPORT

Solar Recommendation Logic

Design Documentation for an Approximate Solar PV System Sizing Engine

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August 2026

Document Type: Technical Design Documentation / SRS Supplement

Document Control

Field	Detail
Title	Solar Recommendation Logic — Technical Documentation
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Document Type	Research Report / Technical Design Documentation
Audience	Developers, QA, Product/Business Stakeholders
Status	Draft — for internal review
Version	1.0

1. Purpose and Overview

1.1 Why This Logic Exists

The Solar Recommendation Logic exists to give a prospective solar customer a fast, self-service, order-of-magnitude estimate of the solar photovoltaic (PV) system that would suit their electricity usage and site, before any professional site survey is performed. It converts a small set of user-supplied inputs (electricity consumption, location, roof area, and optional backup requirements) into an approximate system size, panel count, inverter rating, and — where applicable — battery capacity.

This logic is typically embedded in a website, mobile app, or sales tool as a 'solar calculator' or 'quick quote' feature. It allows a user to understand roughly what a solar installation would look like for their home or business, and gives the sales/technical team a starting point for a more detailed proposal.

1.2 Approximate, Not a Final Engineering Design

It is critical that all stakeholders — technical and non-technical — understand that this logic produces an APPROXIMATE, PRELIMINARY recommendation only. It is explicitly NOT a substitute for:

- A certified site survey (roof orientation, tilt, shading analysis, structural assessment).
- Detailed load/energy audits (appliance-level consumption, load profiles, seasonal variation).
- Local electrical code compliance review and utility interconnection requirements.
- A final engineering design signed off by a licensed solar engineer.

Business rule: Every output generated by this logic must be labeled to the user as an 'approximate estimate' and must carry a disclaimer directing the user toward a professional site assessment before purchase or installation.

2. Required User Inputs

The table below lists every input the recommendation engine consumes, its unit, whether it is mandatory, and its role in the calculation. Where an input is optional, the logic must fall back to a clearly labeled default (see Section 5, Assumptions).

Input	Unit	Required?	Purpose
Monthly electricity consumption	kWh/month	Required (or daily, see next row)	Base value used to derive daily energy demand.
Average daily electricity consumption	kWh/day	Required if monthly not supplied	Alternative/direct base value for sizing.
Available rooftop / installation area	m ²	Required	Upper bound on how many panels can physically be installed; used for feasibility checks.
Location (city/region or coordinates)	text / lat-long	Required	Used to determine Peak Sun Hours (PSH) and regional irradiation, either via lookup table or external solar irradiance API.
Peak Sun Hours (PSH)	hours/day	Optional if location is supplied and a PSH lookup source exists	Represents the equivalent number of full-intensity sun hours per day; directly drives generation calculations.
Solar panel wattage	W (per panel)	Optional (has default)	Used to convert total system kW into number of panels.
System efficiency / performance ratio (PR)	% (0–1)	Optional (has default)	Accounts for real-world losses (wiring, soiling, temperature, inverter, mismatch) versus nameplate capacity.
Battery requirement (yes/no)	boolean	Required if backup is desired	Determines whether battery sizing logic is invoked.
Desired backup duration	hours or days	Required if battery requirement = yes	Drives battery capacity (kWh) calculation.
Grid availability	boolean / category (on-grid, off-grid, hybrid)	Required	Determines system type (grid-tied, off-grid, hybrid) and whether battery is mandatory (off-grid always requires battery).

Configurable parameter: The exact source of Peak Sun Hours (static regional lookup table vs. a live solar irradiance API/NASA POWER/PVGIS-style service) is an implementation decision left to the development team and is NOT fixed by this document.

3. Calculation Logic

This section defines every formula used by the engine. All formulas are mathematical definitions; the specific numeric defaults referenced inside them are documented separately in Section 5 (Assumptions) so that engineering assumptions remain clearly separated from the math itself.

3.1 Daily Energy Consumption (E_{daily})

$$E_{\text{daily}} \text{ (kWh/day)} = \text{Monthly Consumption (kWh)} \div 30$$

If the user supplies daily consumption directly, that value is used as-is and this conversion is skipped.

Variables:

- E_{daily} — average daily electricity consumption, kWh/day
- Monthly Consumption — user-supplied monthly electricity usage, kWh
- 30 — assumed average days per month (configurable; 30.4 may be used for higher precision)

3.2 Required Solar Generation (E_{required})

$$E_{\text{required}} \text{ (kWh/day)} = E_{\text{daily}} \div \text{System Performance Ratio (PR)}$$

This inflates the target generation to compensate for real-world system losses, so that the AC energy actually delivered to the user matches their consumption.

- E_{required} — the DC generation the array must produce per day before losses, kWh/day
- PR — performance ratio / derate factor ($0 < \text{PR} \leq 1$); see Section 5 for default

3.3 Approximate Solar PV System Size (P_{system})

$$P_{\text{system}} \text{ (kW)} = E_{\text{required}} \text{ (kWh/day)} \div \text{Peak Sun Hours (PSH, hours/day)}$$

This is the core sizing formula: it expresses how much installed DC capacity is needed to produce the required daily energy, given the location's average number of peak sun hours.

- P_{system} — recommended solar array capacity, kW (DC, nameplate)
- PSH — Peak Sun Hours for the user's location, hours/day

3.4 Number of Solar Panels (N_{panels})

$$N_{\text{panels}} = \text{CEILING}(P_{\text{system}} \text{ (kW)} \times 1000 \div \text{Panel Wattage (W)})$$

The result is always rounded UP (ceiling) because partial panels cannot be installed.

- N_{panels} — recommended number of panels (integer)
- Panel Wattage — nameplate power of a single panel, W

3.5 Required Rooftop Area (A_{required})

$$A_{\text{required}} \text{ (m}^2\text{)} = N_{\text{panels}} \times \text{Area per Panel (m}^2\text{)}$$

Area per Panel is a configurable default representing the physical footprint of one panel including its frame (see Section 5). This is compared against the user's available rooftop area to flag feasibility (see Section 7, Edge Cases).

3.6 Estimated Battery Capacity (C_{battery})

$$C_{\text{battery}} \text{ (kWh)} = (E_{\text{daily}} \times \text{Backup Duration (days)}) \div (\text{Depth of Discharge (DoD)} \times \text{Battery Efficiency})$$

Applied only when battery requirement = yes, or when the system is off-grid (battery becomes mandatory).

- C_{battery} — usable-to-nameplate battery capacity recommendation, kWh
- Backup Duration — user-desired autonomy, converted to days (e.g., 8 hours = 0.33 days)
- DoD — maximum safe depth of discharge for the battery chemistry ($0 < \text{DoD} \leq 1$); see Section 5
- Battery Efficiency — round-trip efficiency of charge/discharge ($0 < \text{value} \leq 1$); see Section 5

3.7 Approximate Inverter Capacity (P_{inverter})

$$P_{\text{inverter}} \text{ (kW)} = P_{\text{system}} \text{ (kW)} \times \text{Inverter Sizing Factor}$$

The Inverter Sizing Factor is typically slightly below 1.0 (inverters are commonly undersized relative to DC array capacity, a practice called 'DC/AC oversizing') or slightly above 1.0 if surge/starting loads must be covered. The exact factor is a configurable, business-defined value (see Section 5).

4. Recommendation Rules

Recommendation Rules are business decisions that select, round, or bound the raw calculation results from Section 3 into a final, presentable recommendation. They are distinct from the mathematics itself.

Output	Rule
Solar system size (kW)	Take P_system from Section 3.3 and round UP to the nearest 0.1 kW (or nearest commercially available inverter/array step size, e.g., 1 kW increments — configurable business rule).
Number of panels	Take N_panels from Section 3.4 (already rounded up to a whole panel).
Panel capacity	Use the user-supplied panel wattage if provided; otherwise apply the default panel wattage (Section 5).
Inverter size	Take P_inverter from Section 3.7, then round UP to the nearest commercially standard inverter rating (e.g., 3 kW, 5 kW, 8 kW, 10 kW steps — the specific catalog of standard sizes is a configurable business parameter).
Battery size	Take C_battery from Section 3.6 (if applicable), then round UP to the nearest commercially available battery module size (module size is a configurable business parameter, e.g., 2.4 kWh or 5 kWh units).
Estimated installation area	Take A_required from Section 3.5. If A_required exceeds the user's available rooftop area, trigger the 'insufficient area' business rule described in Section 7 instead of silently recommending an oversized array.

Business rule (rounding direction): All physical/discrete quantities (panels, inverter step, battery modules) are rounded UP, never down, so the recommended system always meets or slightly exceeds the calculated requirement rather than under-delivering.

5. Assumptions

The values below are generic, industry-typical defaults used ONLY when the user does not supply a project-specific value, and only where a reasonable industry-standard default exists. They must be clearly surfaced to the user/stakeholder as assumptions, and every one of them should be exposed as a configurable parameter in the application/admin settings rather than hard-coded.

Parameter	Typical Default (assumption)	Notes
Solar panel efficiency	Not directly used in the core formulas above; reflected indirectly via Panel Wattage and Area per Panel	Higher-efficiency panels produce more W in the same physical footprint — affects Area per Panel.
System performance ratio (PR)	0.75 – 0.80 (commonly ~0.77)	Captures cumulative losses: wiring, soiling, mismatch, temperature, inverter conversion, shading.

Parameter	Typical Default (assumption)	Notes
Inverter efficiency	~0.95 – 0.98	Already partially embedded within PR; if modeled separately, must not be double-counted.
Battery round-trip efficiency	~0.90 (Li-ion), ~0.80 (lead-acid)	Chemistry-dependent; must be configurable per battery type offered.
Battery Depth of Discharge (DoD)	0.80 – 0.90 (Li-ion), 0.50 (lead-acid)	Determines usable vs. nameplate battery capacity.
Peak Sun Hours (PSH)	No universal default — MUST be location-specific	Typical global range $\approx$ 3–6.5 hrs/day; sourced from a location lookup table or irradiance API — this data source is a configurable parameter, not invented here.
Panel wattage (default)	550 W (illustrative default for a modern residential monocrystalline panel)	Should be overridden whenever the user or catalog specifies an actual panel model.
Area per panel	~2.6 – 2.8 m ² per panel (illustrative, for large-format modules)	Must match the physical dimensions of the panel model actually being recommended.
Inverter sizing factor (DC:AC ratio)	~1.0 – 1.25 (i.e., inverter AC rating commonly 80–100% of DC array rating)	Business/engineering policy choice; some designs oversize the DC array relative to the inverter.
Temperature / environmental derate	Included within the overall PR default above	In hot climates, a separate temperature derate (e.g., 5–10%) may be modeled explicitly instead of folded into PR — configurable.
Safety / design margin	+10% to +20% applied to E _{required} or final system size (configurable, optional)	A business-level buffer for future consumption growth; not applied unless explicitly enabled.
Days per month (for monthly→daily conversion)	30	30.44 (365/12) may be used for higher precision.

Important: none of the numeric defaults above should be treated as fixed engineering facts for any real project. They are placeholder industry-typical ranges included so the logic is computable and testable; each must be confirmed or replaced with project-specific/manufacturer-specific data before being used for an actual customer proposal.

6. Example Calculation

The following worked example uses realistic, clearly-labeled sample values to demonstrate the full flow from raw consumption to a final recommendation. All input values below are illustrative sample data, not a real customer's data.

6.1 Sample Inputs

Input	Sample Value
Monthly electricity consumption	600 kWh/month
Location	Sample City — assumed PSH = 5 hours/day
Available rooftop area	40 m ²
Panel wattage	550 W (default)
System performance ratio (PR)	0.77 (default)
Area per panel	2.7 m ² (default)
Grid availability	On-grid, hybrid with battery backup desired
Desired backup duration	6 hours (0.25 day)
Battery DoD	0.85 (Li-ion default)
Battery round-trip efficiency	0.90 (Li-ion default)
Inverter sizing factor	1.1 (default)

6.2 Step-by-Step Calculation

Step 1 — Daily energy consumption

$$E_{\text{daily}} = 600 \div 30 = 20 \text{ kWh/day}$$

Step 2 — Required solar generation

$$E_{\text{required}} = 20 \div 0.77 \approx 25.97 \text{ kWh/day}$$

Step 3 — Approximate PV system size

$$P_{\text{system}} = 25.97 \div 5 \approx 5.19 \text{ kW}$$

Step 4 — Number of panels

$$N_{\text{panels}} = \text{CEILING}(5.19 \times 1000 \div 550) = \text{CEILING}(9.44) = 10 \text{ panels}$$

Step 5 — Required rooftop area

$$A_{\text{required}} = 10 \times 2.7 = 27 \text{ m}^2$$

Feasibility check: $A_{\text{required}} (27 \text{ m}^2) \leq \text{Available rooftop area} (40 \text{ m}^2) \rightarrow \text{PASS}$. The recommendation can proceed without a space-constraint warning.

Step 6 — Battery capacity (backup requested)

$$C_{\text{battery}} = (20 \times 0.25) \div (0.85 \times 0.90) = 5 \div 0.765 \approx 6.54 \text{ kWh}$$

Step 7 — Inverter capacity

$$P_{\text{inverter}} = 5.19 \times 1.1 \approx 5.71 \text{ kW}$$

6.3 Final Rounded Recommendation (applying Section 4 rules)

Output	Raw Value	Recommended (rounded)
PV system size	5.19 kW	5.2 kW
Number of panels	9.44 → 10	10 panels
Panel wattage	—	550 W each
Rooftop area required	27 m ²	≈ 27 m ² (of 40 m ² available)
Inverter size	5.71 kW	6 kW (nearest standard size, illustrative)
Battery capacity	6.54 kWh	7.2 kWh (nearest standard module size, illustrative)

7. Edge Cases and Validation

Scenario	Expected Handling
Very low electricity consumption (e.g., near-zero kWh)	Flag as 'below minimum viable system size'; recommend a minimum catalog system (configurable floor, e.g., smallest available inverter/panel combination) rather than an unrealistically tiny fractional system.
Very high electricity consumption	Flag for manual/commercial-scale review if P_system exceeds a configurable residential ceiling (e.g., a large kW threshold); route to a commercial sizing workflow or sales engineer instead of returning an automated number.
Insufficient rooftop area (A_required > available area)	Do not silently shrink the system. Recommend the maximum system that DOES fit the available area, clearly show the resulting energy shortfall versus the user's actual consumption, and label the result as 'partial coverage — reduced by roof constraint'.
Missing or invalid inputs (e.g., no consumption value, no location)	Reject the calculation and return a validation error identifying exactly which required field is missing; do not silently substitute a default for a mandatory field such as consumption.
Zero or negative values (consumption, area, wattage, PSH, etc.)	Reject as invalid input with a specific error message; zero/negative values are physically meaningless and must not be passed into the formulas.
Unusually high backup requirement (e.g., multi-day autonomy)	Allow the calculation to proceed but flag the resulting battery size as 'large — verify feasibility and cost with a specialist,' since very large battery banks have space, weight, and cost implications beyond this tool's scope.
Different panel wattages supplied by user/catalog	Formulas in Section 3 are wattage-agnostic — simply substitute the supplied wattage; N_panels and A_required will scale accordingly. No formula change needed.
Locations with different solar irradiation (PSH)	PSH must be looked up per location (see Section 2 note); the same formulas apply globally, but P_system, N_panels, and A_required will

Scenario	Expected Handling
	vary directly with PSH. A location with no PSH data available should trigger a 'missing input' error rather than assuming a default PSH.

8. Limitations

This logic deliberately trades precision for speed and accessibility. The following factors are NOT modeled and require a professional site survey or detailed engineering analysis before any purchase, permitting, or installation decision:

- Roof orientation, tilt angle, and true shading analysis (trees, adjacent buildings, chimneys) across the day and across seasons.
- Structural assessment of whether the roof can bear the panel/mounting load.
- Detailed, appliance-level or time-of-use load profiling (this logic uses only an averaged daily/monthly figure).
- Seasonal variation in both consumption and solar generation (the logic uses a single averaged PSH and a single averaged daily consumption).
- Local electrical code, permitting, and utility interconnection/net-metering rules.
- Actual pricing, incentives, financing, and payback/ROI analysis.
- Detailed electrical design: wiring gauge, protection devices, string sizing, voltage drop, and safety compliance.
- Manufacturer-specific derating curves, warranty terms, and degradation over the system's lifetime.

The recommendation should always be presented to the end user together with a clear statement that it is a preliminary estimate, and that final system design and pricing require a site visit and a licensed solar professional's sign-off.

9. Pseudo-Code

The pseudocode below is implementation-language-independent and mirrors the logic and rules described in Sections 2–7.

```

FUNCTION recommendSolarSystem(inputs, config):

    // ---- 1. VALIDATE INPUTS ----
    IF inputs.monthlyConsumption IS MISSING AND inputs.dailyConsumption IS MISSING:
        RETURN error("Consumption is required")
    IF inputs.location IS MISSING:
        RETURN error("Location is required")
    IF inputs.rooftopArea IS MISSING:
        RETURN error("Rooftop area is required")
    FOR EACH numeric field IN [monthlyConsumption, dailyConsumption, rooftopArea,
panelWattage, PSH]:
        IF field IS PROVIDED AND field <= 0:
            RETURN error(field + " must be a positive number")

    // ---- 2. RESOLVE DEFAULTS (config-driven, see Section 5) ----
    PR                = inputs.performanceRatio OR config.defaultPR
    panelWattage      = inputs.panelWattage OR config.defaultPanelWattage
    areaPerPanel      = config.defaultAreaPerPanel
    inverterFactor    = config.defaultInverterSizingFactor
    PSH              = inputs.PSH OR lookupPSH(inputs.location, config.irradianceSource)
    IF PSH IS MISSING:
        RETURN error("Peak Sun Hours unavailable for this location")

    // ---- 3. DAILY CONSUMPTION ----
    E_daily = inputs.dailyConsumption OR (inputs.monthlyConsumption /
config.daysPerMonth)

    // ---- 4. CORE CALCULATIONS (Section 3) ----
    E_required = E_daily / PR
    P_system   = E_required / PSH
    N_panels   = CEILING( (P_system * 1000) / panelWattage )
    A_required = N_panels * areaPerPanel
    P_inverter = P_system * inverterFactor

    // ---- 5. AREA FEASIBILITY CHECK ----
    warnings = []
    IF A_required > inputs.rooftopArea:
        maxPanelsThatFit = FLOOR(inputs.rooftopArea / areaPerPanel)
        N_panels        = maxPanelsThatFit
        P_system        = (N_panels * panelWattage) / 1000
        A_required      = N_panels * areaPerPanel
        warnings.add("Recommendation reduced to fit available roof area; " +
                    "generation will not fully cover consumption.")

    // ---- 6. BATTERY SIZING (optional) ----
    C_battery = NULL

```

```

IF inputs.batteryRequired OR inputs.gridAvailability == "off-grid":
    backupDays = convertToDays(inputs.backupDuration)
    DoD        = config.defaultDoD
    battEff    = config.defaultBatteryEfficiency
    C_battery  = (E_daily * backupDays) / (DoD * battEff)
    IF C_battery > config.largeBatteryThreshold:
        warnings.add("Battery size is large; verify feasibility with a specialist.")

// ---- 7. RANGE / SCALE CHECKS ----
IF P_system < config.minResidentialSystemKW:
    P_system = config.minResidentialSystemKW
    warnings.add("Consumption is very low; minimum catalog system applied.")
IF P_system > config.maxResidentialSystemKW:
    warnings.add("System size exceeds typical residential range; route to commercial
review.")

// ---- 8. APPLY BUSINESS ROUNDING RULES (Section 4) ----
P_system_final = roundUpToStep(P_system, config.systemSizeStep)
inverter_final = roundUpToCatalog(P_inverter, config.inverterCatalog)
battery_final  = C_battery IS NOT NULL ?
                roundUpToCatalog(C_battery, config.batteryModuleSize) : NULL

// ---- 9. RETURN RECOMMENDATION ----
RETURN {
    pvSystemSizeKW: P_system_final,
    numberOfPanels: N_panels,
    panelWattageW: panelWattage,
    inverterSizeKW: inverter_final,
    batteryCapacityKWh: battery_final,
    areaRequiredM2: A_required,
    assumptionsUsed: { PR, PSH, areaPerPanel, inverterFactor, DoD, battEff },
    warnings: warnings,
    disclaimer: "Approximate estimate only. Confirm with a professional site
survey."
}

```

10. Decision Flow

A concise, linear decision flow developers can map directly onto application/UI logic:

- 1. Collect inputs: consumption (monthly or daily), location, rooftop area, grid availability, and — if backup is desired — battery flag and backup duration.
- 2. Validate: reject missing mandatory fields; reject zero/negative numeric values; return field-specific error messages.
- 3. Resolve defaults: fill any optional missing input (panel wattage, PR, area per panel, inverter factor, DoD, battery efficiency) from configuration; look up PSH by location if not supplied.
- 4. If PSH cannot be resolved for the location, stop and return an error rather than guessing.

- 5. Compute $E_{\text{daily}} \rightarrow E_{\text{required}} \rightarrow P_{\text{system}} \rightarrow N_{\text{panels}} \rightarrow A_{\text{required}} \rightarrow P_{\text{inverter}}$, in that order (each depends on the previous).
- 6. Compare A_{required} against available rooftop area. If it does not fit, recompute a reduced system that does fit and attach a coverage-shortfall warning.
- 7. If battery is required (explicitly requested, or system is off-grid), compute C_{battery} from backup duration, DoD, and battery efficiency; flag if unusually large.
- 8. Apply scale checks: enforce a minimum viable system size; flag systems above the typical residential ceiling for manual/commercial review.
- 9. Apply business rounding rules to convert raw numbers into catalog-realistic values (system size step, inverter catalog sizes, battery module sizes).
- 10. Assemble the final output object (Section 11), including every assumption used and every warning raised, and always attach the 'approximate estimate' disclaimer.
- 11. Return the recommendation to the calling application/UI for display to the user.

11. Output Specification

The recommendation engine's output must be a structured object containing at least the following fields:

Field	Type / Unit	Description
pvSystemSizeKW	number, kW	Final recommended PV array capacity.
numberOfPanels	integer	Recommended panel count.
panelWattageW	number, W	Wattage of the panel model used in the calculation.
inverterSizeKW	number, kW	Recommended inverter capacity.
batteryCapacityKWh	number, kWh (nullable)	Recommended battery capacity; null/omitted if no battery requested and system is on-grid.
areaRequiredM2	number, m ²	Approximate rooftop area needed for the recommended panel count.
assumptionsUsed	object	Every default value actually applied (PR, PSH, area per panel, inverter factor, DoD, battery efficiency, etc.), so the result is fully auditable.
warnings	array of strings	Any edge-case flags raised (e.g., area shortfall, oversized battery, above/below typical range).
disclaimer	string	Fixed, mandatory text stating the result is an approximate estimate requiring professional site verification.

All numeric outputs should be expressed consistently in kWh, kW, W, kWh/day, and m², matching the units used throughout this document, so the UI, backend, and any downstream reporting stay aligned.

— *End of Document* —